Huawei Access Node (YEL-ACC-52) Standard Operating Procedure for Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss)

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6439

Description

This failure domain occurs on the Huawei Access Node YEL-ACC-52 due to Optical Transport Failure in the SFP/XFP port. This type of failure can be caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The Huawei Access Node is a part of the Optical Transport Network (OTN) that transmits data over fiber-optic cables. The OTN uses wavelength division multiplexing (WDM) to transmit multiple signals over a single fiber, each on a different wavelength. The SFP/XFP ports on the Huawei Access Node are responsible for transmitting and receiving these signals. When a SFP/XFP degrades, it can cause a loss of signal strength, leading to a failure in the OTN. Similarly, fiber cuts or Rx/Tx power loss can also cause a failure in the OTN. In Huawei equipment, Optical Transport Failures are typically reported by the NE (Network Element) as an Alarm Code ALARM_6439, indicating a Warning severity. This SOP outlines the steps to diagnose and resolve the issue.

Pre-requisite Checks

- Ensure that the Yelahanka Sub-station is accessible to dispatch remote hands.

- Verify the status of the redundant link to ensure it is not causing the failure.

- Check the environmental conditions at the Yelahanka Sub-station to rule out any external factors causing the failure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface, Expected Output: Interface SFP1/0/1 is down due to SFP degradation, Description: This command checks the status of the optical transport interface and displays the reason for the failure. The interface status and reason are checked in the control plane, which helps isolate the issue.

Command: display optical-transport interface statistics, Expected Output: Rx power is -30 dBm and Tx power is -20 dBm, Description: This command checks the statistics of the optical transport interface and
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displays the receive and transmit power levels. This helps verify if
there is a power loss issue causing the failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transport interface SFP1/0/1, Description: This command replaces the SFP/XFP module on the optical transport interface. The new SFP/XFP module is inserted into the interface, and the optical transport interface is reinitialized to ensure proper operation.

Step 2: Step: Verify the fiber connections, Command: display optical-transport interface fiber connections, Description: This command checks the fiber connections of the optical transport interface and verifies that the fiber is properly connected and not cut.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust optical-transport interface power levels, Description: This command adjusts the receive and transmit power levels of the optical transport interface to ensure that the signal strength is within the recommended range.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output:
Interface SFP1/0/1 is up and operational, Description: This command
checks the status of the optical transport interface and verifies that
it is up and operational.

Command: display optical-transport interface statistics, Expected
Output: Rx power is -10 dBm and Tx power is -5 dBm, Description: This
command checks the statistics of the optical transport interface and
verifies that the receive and transmit power levels are within the
recommended range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore optical-transport interface SFP1/0/1, Description: This command restores the original SFP/XFP module on the optical transport interface. The original SFP/XFP module is reinserted into the interface, and the optical transport interface is reinitialized to ensure proper operation.

- Step: Verify the fiber connections, Command: display optical-transport interface fiber connections, Description: This command checks the fiber connections of the optical transport interface and verifies that the fiber is properly connected and not cut.